

Data Structures & Algorithms

Problem Set 1 • Counting Work • Asymptotics

Due: Friday, Feb 20, 2026 • 11:59 PM (ET)

PS 1

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What we're practicing: translating loops into summations, converting summations to closed forms, and comparing growth rates using $O(\cdot)$, $\Omega(\cdot)$, and $\Theta(\cdot)$.

INSTRUCTIONS

- Assume `Work()` runs in constant time, with cost $c > 0$ per call, independent of input.
- For "closed form," give an exact expression (not just $O(\cdot)$) unless the prompt says otherwise.
- Show your reasoning: summations are fine, but don't stop at a raw double-sum.
- Logarithms are base 2 unless stated otherwise (the base doesn't matter for asymptotics).

Question 1 — Counting Nested Loop Work

15 pts

Let $T(n)$ be the total cost contributed by calls to `Work()` in the algorithm below. Derive a closed form for $T(n)$ (exact, in terms of n and c), and give the tight $\Theta(\cdot)$ bound.

```
function B1(n: positive integer):
  total ← 0
  for i from 1 to n by multiplying by 4 do
    for j from 1 to i do
      total ← total + Work()
    end for
  end for
  return total
end function
```

Hint: the outer loop visits $i = 1, 4, 16, \dots$ up to n . Convert to a geometric series of partial sums.

Question 2 — Return Value as a Closed Form

20 pts

The following function returns a number (not a runtime). Derive the exact value returned, as a function of n . You may assume n is a positive integer.

```
function B2(n: positive integer):
  s ← 0
  for i from 1 to n do
    for j from 1 to (2i - 1) do
      s ← s + 1
    end for
  end for
  return s
end function
```

Hint: recognize the inner loop length and use a known summation identity.

Question 3 — True/False on Asymptotic Claims

3 × 6 pts

For each statement, mark it **True** or **False**. Provide a short justification: either a proof sketch (e.g., limits, inequalities) or a counterexample.

a. $n^3 = O(2^n)$

b. $\frac{\log n}{\log \log n} = O(\sqrt{\log n})$

c. $n \log n = \Omega(n^{1.1})$

Tip: if you use limits, explicitly state what you're taking the limit of and why it implies the bound.

Question 4 — Ordering by Growth Rate + A Property Proof

25 pts

(a) Sort the following functions from slowest growth to fastest growth (increasing order). If two functions are $\Theta(\cdot)$ of each other, group them together and briefly justify.

$$f_1(n) = n^{2.5}, \quad f_2(n) = 3^n, \quad f_3(n) = n^2 \log n, \quad f_4(n) = 2^{n/3} \cdot n^7, \quad f_5(n) = (\log n)^8$$

(b) Suppose $f(n)$ and $g(n)$ are functions that are eventually positive (there exists n_0 such that for all $n \geq n_0$, $f(n) > 0$ and $g(n) > 0$). Prove:

$$\min\{f(n), g(n)\} = \Theta\left(\frac{f(n)g(n)}{f(n) + g(n)}\right)$$

(c) Give an explicit example of functions f, g (eventually positive) such that neither $f = O(g)$ nor $g = O(f)$ holds. Briefly justify.

Tip for (b): bound the expression between constant multiples of $\min\{f, g\}$ using simple inequalities like $f + g \leq 2 \max\{f, g\}$ and $f + g \geq \max\{f, g\}$.